1. Run report: EXP-CMFB-cea1d2

Observations only. This report describes what the eight frozen trials ran and what they recorded. It states no verdict, no support or weakening, and no significance; those belong to an independent evidence review. A recorded field that the frozen rule computed mechanically is quoted as recorded, not adopted. Numbers are marked *recorded* (read from a run file) or *derived* (arithmetic on recorded values, shown).

1.1. Identity

Experiment	EXP-CMFB-cea1d2 (hypothesis H-CMFB-9b2eed, question RQ-CMFB-33610e, idea IDEA-20261015-cba035)
Executor task / claim	TASK-20261015-1c0251, claim executor-run-cmfb epoch 1 (commit cc2ece9a49)
Frozen plan	trial-plan.json sha256 d580d0503c2b7c9e4437d1f03eca2d6a93d86d7620e81ccb11c4a7a3b3ff50fe
Code	commit cc2ece9a497e805ec8ac76f38a5b28a3549ac08e, dirty = false (recorded in manifest)
Host	Linux x86_64, 4 CPUs, 16.9 GB RAM, Python 3.13.16, numpy 2.5.3 (recorded)
Window	2026-10-19 05:41:51Z to 05:57:50Z, wall about 16 min (recorded start and finish times)
Instance list	sha256 38cf55093614d8f8761e392d75e17ae5f0266e50b69b7abd58d6f01c3b2fa314 (identical in the pre-run inventory), 107 members total (recorded)

1.2. Trials

Requested 8, completed 8, failed 0, timed out 0. Every trial's checker returned success and the supervisor recorded `output_validated`. Run directories are under `experiments/EXP-CMFB-cea1d2/runs/`.

Trial	Run ID	Started (UTC)	Duration
t1 inventory	RUN-CMFB-2373be	05:41:51	2.7 s
t2 gate: instrument	RUN-CMFB-5905c0	05:41:56	24.4 s
t3 gate: sensitivity	RUN-CMFB-66b0fe	05:42:23	66.5 s
t4 measure C16 + C20	RUN-CMFB-3dc120	05:43:33	60.6 s
t5 measure C24, FAM	RUN-CMFB-42a578	05:44:36	241.9 s
t6 measure C24, N-A	RUN-CMFB-e13f0e	05:48:40	253.7 s
t7 measure C24, N-B	RUN-CMFB-c71d35	05:52:57	288.3 s
t8 analysis	RUN-CMFB-2010d3	05:57:48	2.2 s

Durations are *derived* from the recorded start and finish times. Peak memory of the analysis trial: 58.3 MB (recorded).

1.3. Instances and cells

Recorded family-member counts: C16 = 4, C20 = 10, C24 = 13. Recorded arm sizes (FAM / N-A / N-B): C16 4 / 8 / 8, C20 10 / 16 / 16, C24 13 / 16 / 16. The pooling rule P1 was recorded as not applied. The pre-registered Bateman-Horn expectations were 3.11, 8.01 and 14.42; exact counts over expectation were 1.28, 1.25 and 0.90 (recorded).

The analysis recorded `powered` for C24 only; `mde_ok` was recorded false for C16 and C20, and `growth_assessable` = false (one powered cell). Recorded C20 arm MDEs were FAM 0.243 %, N-A 0.268 %, N-B 0.245 %, against the contract threshold 0.25 %.

1.4. Gates and controls (recorded)

- Instrument gate t2 and sensitivity gate t3 passed; invalidation rules I1 to I9 all recorded true. The run is recorded `official` and `all_inputs_official`.
- qr_window positive control (f(x) residuosity): pooled scan-efficiency ratio R between 1.9941 and 2.0009 in all 9 pools, all inside the recorded band 1.98 to 2.02. Negative control (Legendre of x), R_wrong: 0.9970 to 1.0006 in all 9 pools. See the right panel of the figure.
- Planted-base sensitivity: detected 7 of 8 (C16), 8 of 8 (C20), 8 of 8 (C24). Pooled recorded ratio against the known value: C16 1.02667 vs 1.03035, C20 1.01091 vs 1.01247, C24 1.00525 vs 1.00521, all recorded within tolerance; pooled randomization p = 5.0e-5 in each cell.

1.5. Measured coverage excess

T is the recorded mean over members of (coverage ratio against 200 matched random bases) minus 1, in percent. Intervals are the recorded 95 % moving-block-bootstrap intervals. Holm p is recorded only for powered cells. The last two columns show the recorded variance ratio and the same ratio rescaled (see below).

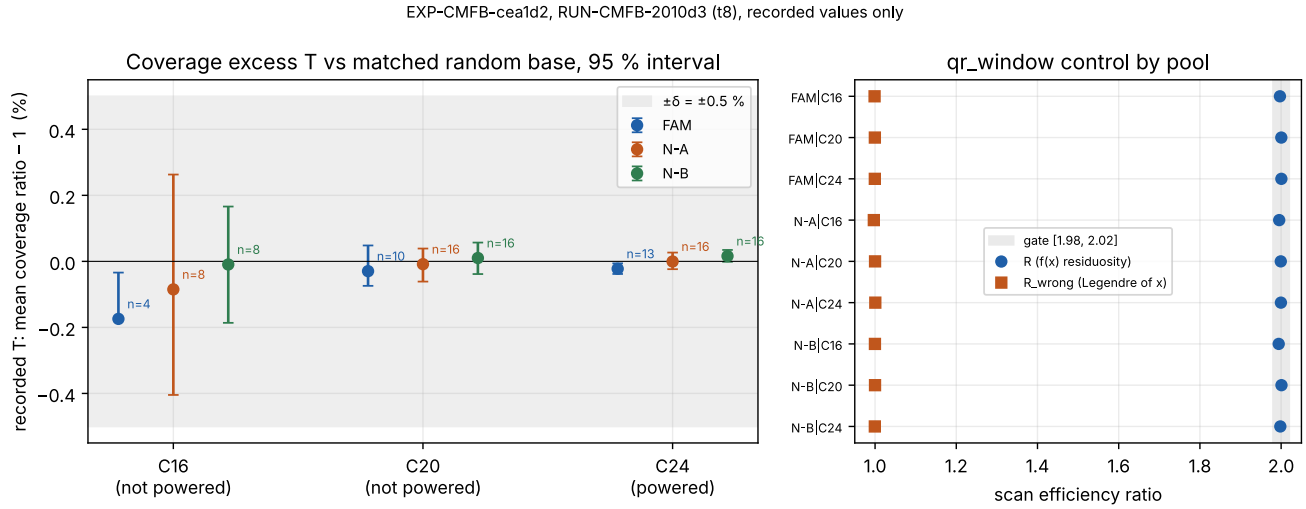


Figure 1: Left: recorded T with 95 % intervals per arm and cell (n as labeled), grey band = the contract resolution ±0.5 %. Right: recorded scan-efficiency ratios per pool. Source: RUN-CMFB-2010d3 raw-result.json; script plot.py. Missing arm-cells: none.

Arm	Cell	n	T (%)	95 % interval (%)	Holm p	rand. p	MDE (%)	meas/formula	derived
FAM	C16	4	−0.174	[−0.174, −0.034]	-	0.411	1.332	2.56	0.90
FAM	C20	10	−0.030	[−0.074, +0.048]	-	0.441	0.243	2.57	0.90
FAM	C24	13	−0.022	[−0.038, −0.006]	0.094	0.031	0.065	2.68	0.94
N-A	C16	8	−0.085	[−0.405, +0.263]	-	0.686	1.318	2.39	0.84
N-A	C20	16	−0.009	[−0.061, +0.039]	-	0.843	0.268	2.58	0.90
N-A	C24	16	−0.000	[−0.024, +0.027]	0.969	0.969	0.063	2.64	0.92
N-B	C16	8	−0.009	[−0.186, +0.166]	-	0.969	1.537	2.47	0.86
N-B	C20	16	+0.010	[−0.038, +0.057]	-	0.795	0.245	2.72	0.95
N-B	C24	16	+0.016	[−0.001, +0.035]	0.159	0.079	0.057	2.77	0.97

All nine recorded 95 % intervals lie inside ±0.5 %. Recorded Holm adjusted p-values for the three powered-cell tests: FAM 0.094, N-A 0.969, N-B 0.159; `holm_rejection_below_delta` is empty. Recorded contrasts in C24 (FAM minus N-A, FAM minus N-A near-family stratum, N-A minus N-B) have 99 % intervals that include zero. The lead tier is recorded null.

The analysis recorded the field `classification` = `NULL_AT_RESOLUTION`, computed mechanically from the frozen decision rule. It is quoted here as recorded; it is not a review decision, and the contract states it is reported at resolution 0.5 %, not as “no effect”.

1.6. Departures and defects observed

- **Variance check (I10) fires in all 9 arm-cells.** The recorded `sigma_flag_I10` is true everywhere, with recorded measured/formula ratios 2.39 to 2.77. The contract was revised before freezing to centre this check on 1.4/√N (band 0.75x to 1.33x), but `analyze.py` still computes the Poisson-occupancy formula (`count.sigma_rel_formula`), the source of the earlier 0.49/√N constant. *Derived* (approximate): multiplying each recorded ratio by 0.49/1.4 = 0.35 gives 0.84 to 0.97, inside the contract band. I10 is a non-void flag; the run is not voided, but the flag text recorded in the tables reflects the code, not the contract.
- **Powered cells.** One cell (C24) was recorded powered; C20 failed the MDE condition because the N-A arm’s MDE (0.268 %) exceeded 0.25 %. The contract stated before the run that no slope claim was possible.
- **FAM at C16.** The recorded interval for FAM C16 has T at its lower end (T = −0.174 %, interval [−0.174, −0.034] %, n = 4); recorded as is, no explanation computed here.
- The checker and the supervisor recorded no failed or retried trial. No data is missing.

1.7. What this report does not do

It does not review the run, set any hypothesis status, or say whether the null reading is warranted. Reproduce the figure with `python3 plot.py` in this directory (reads RUN-CMFB-2010d3 only).